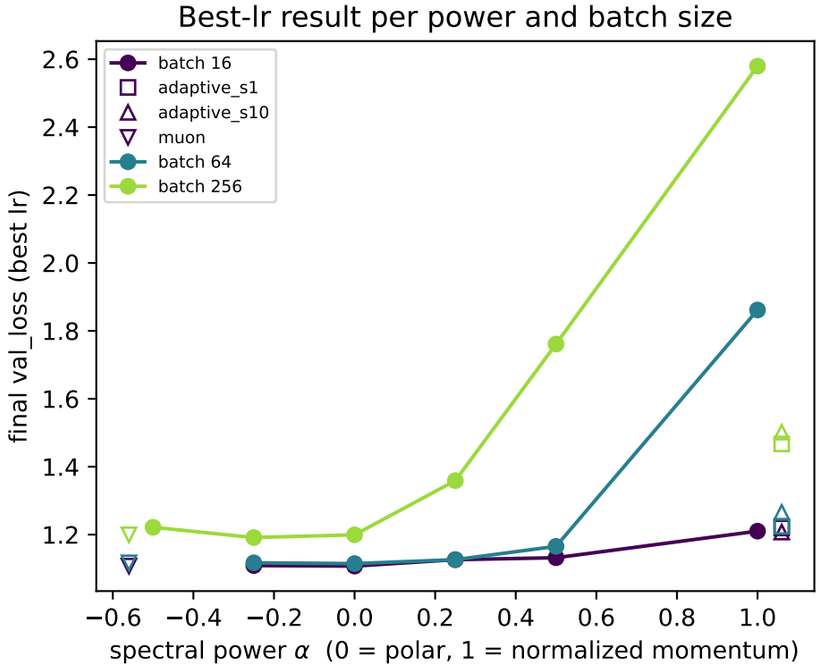
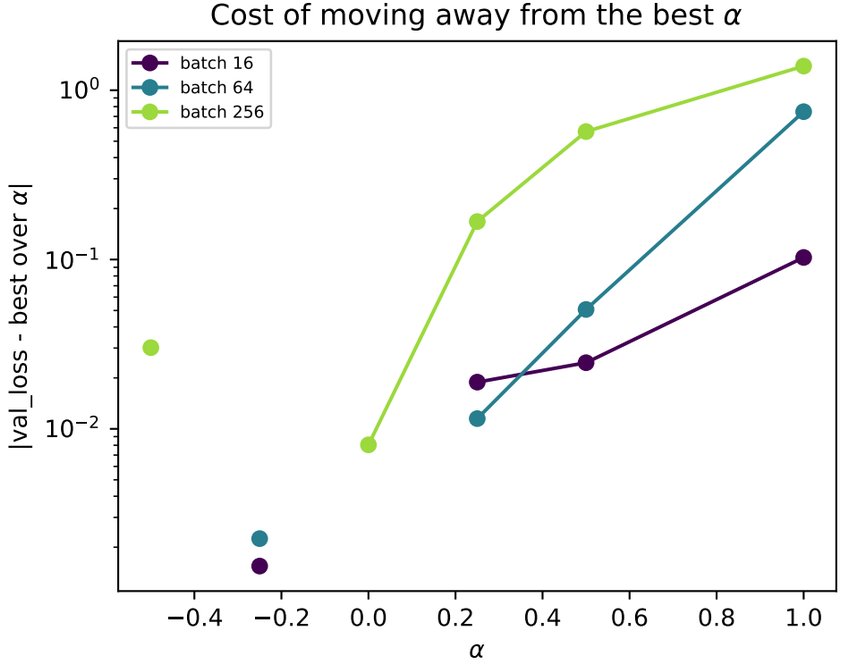
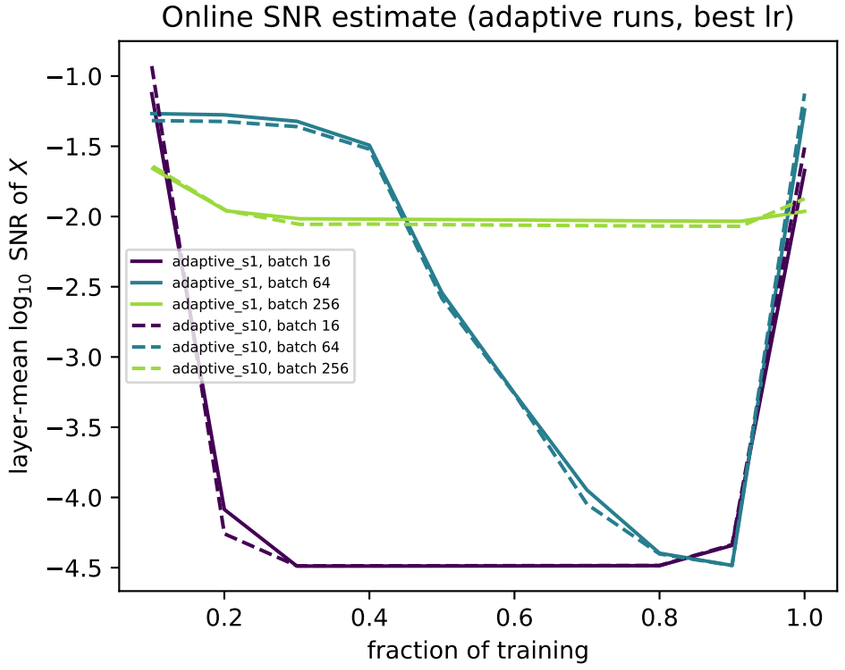

Best-lr result per power and batch size
batch 16
adaptive_s1
adaptive_s10
muon
batch 64
batch 256
final val_loss (best lr)
spectral power α (0 = polar, 1 = normalized momentum)


Cost of moving away from the best α
batch 16
batch 64
batch 256
|val_loss - best over α|
α


Online SNR estimate (adaptive runs, best lr)
adaptive_s1, batch 16
adaptive_s1, batch 64
adaptive_s1, batch 256
adaptive_s10, batch 16
adaptive_s10, batch 64
adaptive_s10, batch 256
layer-mean log10 SNR of X
fraction of training